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Ministry of Higher Education and Scientific Research

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Faculty of Sciences
Department of Computer Science

Bachelor's Report
In Computer Science

Specialty:
Information Systems and Software Engineering (ISIL)
Computer Systems (SIQ)

Web-Based Recruitment System with AI-Powered Resume Matching

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Abstract

This project presents the design and implementation of an intelligent AI-based recruitment management platform for Algérie Télécom. The platform automates the recruitment process through semantic CV analysis, intelligent candidate matching, application tracking, and HR management tools. Developed using Django and modern web technologies, the system integrates artificial intelligence techniques based on SentenceTransformers and cosine similarity to improve candidate evaluation and recruitment efficiency.

Keywords: artificial intelligence, recruitment management, semantic matching,candidate analysis.

Résumé

Ce projet présente la conception et la réalisation d'une plateforme intelligente de gestion du recrutement basée sur l'intelligence artificielle pour Algérie Télécom. La plateforme automatise le processus de recrutement grâce à l'analyse sémantique des CV, au matching intelligent des candidats, au suivi des candidatures et aux outils de gestion des ressources humaines. Développé avec Django et des technologies web modernes, le système intègre des techniques d'intelligence artificielle basées sur SentenceTransformers et la similarité cosinus afin d'améliorer l'évaluation des candidats et l'efficacité du recrutement.

Mots-clés : intelligence artificielle, gestion du recrutement, matching sémantique, analyse des candidats.

الملخص

يهدف هذا المشروع إلى تصميم وتطوير منصة ذكية لإدارة التوظيف تعتمد على الذكاء الاصطناعي لصالح مؤسسة اتصالات الجزائر. تقوم المنصة بأتمتة عملية التوظيف من خلال التحليل الدلالي للسير الذاتية، والمطابقة الذكية بين المترشحين والمناصب، وتتبع طلبات التوظيف، بالإضافة إلى أدوات إدارة الموارد البشرية. تم تطوير النظام باستخدام Django وتقنيات الويب الحديثة، مع دمج تقنيات الذكاء الاصطناعي المعتمدة على SentenceTransformers وخوارزمية التشابه الكوني من أجل تحسين تقييم المترشحين ورفع كفاءة عملية التوظيف.

الكلمات المفتاحية: الذكاء الاصطناعي، إدارة التوظيف، المطابقة الدلالية، تحليل المترشحين.

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List of Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
UML	Unified Modeling Language
CSRF	Cross-Site Request Forgery
DRF	Django REST Framework
JWT	JSON Web Token
MCQ	Multiple Choice Questions
PDF	Portable Document Format
SQL	Structured Query Language
HR	Human Resources

General Introduction

1.1 Context

Algérie Télécom is the leading telecommunications operator in Algeria, providing fixed-line telephony, high-speed internet, and data transmission services nationwide. As a major public company with thousands of employees, it receives a high volume of job applications for various technical and administrative positions.

Currently, the recruitment process at Algérie Télécom is managed manually. HR managers face several challenges: processing hundreds of CVs per offer, ensuring objective candidate evaluation, reducing response time, and tracking application status efficiently.

These challenges highlight the need for an intelligent, automated recruitment system capable of analyzing CVs semantically, ranking candidates objectively, and streamlining the entire recruitment workflow. This project aims to fulfill this need by developing an AI-powered recruitment management platform tailored to Algérie Télécom.

1.2 Problem Statement

Manual candidate selection requires significant time and effort from HR managers.

1.2.1 Traditional Recruitment Systems

Traditional recruitment systems rely on manual CV screening and keyword-based searches. These systems present several limitations:

- Time-consuming manual review of applications.
- Subjective candidate evaluation.
- Lack of semantic understanding.
- No automated ranking mechanism.

1.2.2 Limitations of Existing Systems

Based on the comparative analysis, the main limitations of existing systems are:

- No integrated technical testing module.
- Absence of AI chatbot for candidate assistance.
- Limited automation of the recruitment workflow.

The recruitment process at Algérie Télécom faces two main challenges.

First, recruitment management lacks centralized data tracking, automated workflows, and communication tools.

Second, the candidate matching system relies on simple keyword searches without semantic understanding.

As a result, the process suffers from delays, lack of traceability, and missed qualified candidates.

1.2.3 Positioning of Our Contribution

To address these limitations, our system proposes:

- Semantic CV matching using SentenceTransformers and cosine similarity.
- Integrated online technical testing module.
- AI-powered chatbot for candidate guidance.
- Complete automation of the recruitment workflow with centralized data tracking.
- Objective candidate ranking based on AI scoring.

These improvements provide an intelligent platform combining centralized management with AI-powered semantic matching.

1.3 Research Questions

This project seeks to answer the following research questions:

- How can artificial intelligence improve recruitment management?
- How can semantic analysis improve candidate-job matching?
- How can the recruitment process be automated efficiently?

1.4 Objectives

The main objectives of the project are:

1.4.1 Centralize Recruitment Management

The first objective is to create a centralized platform where all recruitment data is stored in a single database. This includes job offers, candidate applications, CVs, test results, and communication history. Centralization eliminates scattered information across emails and spreadsheets, allowing HR managers to access and manage all recruitment activities from one unified interface.

1.4.2 Automate CV Analysis

The second objective is to automate the extraction and analysis of information from candidate CVs. Instead of manually reading each CV, the system will automatically extract key information such as skills, education, work experience, certifications, and contact details. This automation significantly reduces the time HR managers spend on initial candidate screening.

1.4.3 Implement Intelligent Candidate Matching Using Semantic Analysis

The third objective is to develop an AI-powered matching engine that goes beyond simple keyword searches. Using semantic analysis and natural language processing (NLP), the system will understand the meaning and context of text in both job descriptions and CVs. This allows the system to identify relevant candidates even when different words or expressions are used, ensuring more accurate and intelligent matching.

1.4.4 Provide Interactive Dashboards for Candidates and HR Managers

The fourth objective is to design user-friendly dashboards tailored to each user type. Candidates will have a dashboard to track their application status, view test invitations, and receive feedback. HR managers will have a dashboard to monitor all applications, view AI-generated scores, filter candidates, and manage the recruitment workflow. These dashboards improve transparency and user experience.

1.4.5 Improve Recruitment Decision-Making Through Data-Driven Insights

The fifth objective is to support HR managers with data-driven insights. The system will provide AI-generated scores, candidate rankings, skill gap analysis, and recruitment analytics. These insights help HR managers make objective, consistent, and informed hiring decisions rather than relying on subjective judgment alone.

1.5 Report Organization

This thesis is structured into four main chapters.

Chapter 1 presents the context, problem statement, research questions, and objectives.

Chapter 2 provides a comprehensive review of existing literature on intelligent recruitment systems, CV/job matching techniques, and chatbots.

Chapter 3 covers the system design, including requirements specification, UML modeling, and the AI matching system.

Chapter 4 describes the technical implementation, development environment, technologies used, application interfaces, and test results.

Finally, a general conclusion summarizes the contributions, identifies limitations, and proposes future perspectives.

System Design and Modeling

2.1 Introduction

Analysis and design are essential phases in the development of a software system. This chapter presents the architecture of our platform through UML diagrams and the description of main functionalities.

2.2 Requirements Specification

2.2.1 Actors Identification

The platform involves three main types of actors:

- **Candidate:** External user who consults job offers, submits applications, takes technical tests, and tracks application status.
- **HR Manager (Recruiter):** Internal manager who publishes offers, reviews applications, manages tests, and makes final decisions.
- **Administrator:** Technical manager of the platform, responsible for system configuration and user account management.

2.2.2 Functional Requirements

Table 2.1: Role Assignment by Actor

Actor	**Role**
Candidate	• Create an account and manage their profile. • Consult available job offers. • Submit an application with CV upload (PDF format). • Take online technical tests. • Track application status. • Interact with the AI chatbot.
HR Manager	• Publish and manage job offers. • Consult and filter received applications. • Review technical test results. • Validate or reject applications. • Manage the recruitment dashboard.
Administrator	• Manage user accounts. • Configure platform settings. • Monitor system logs.

2.3 System Analysis and Design Using UML Diagrams

2.3.1 Use Case Diagrams

Use case diagrams provide a graphical overview of interactions between actors and the system. They help identify main functionalities and their relationships.

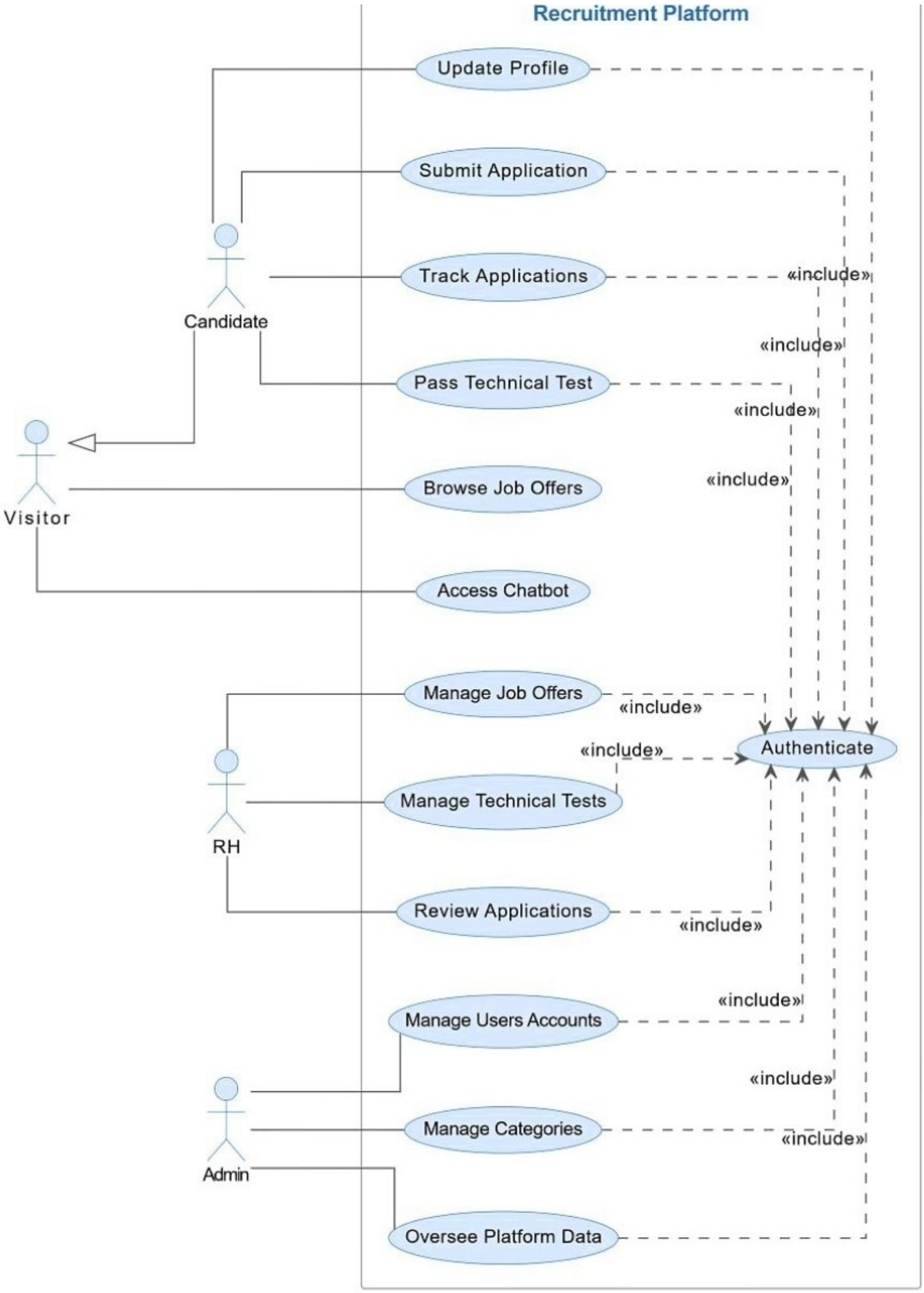

Figure 2.1: Global Use Case Diagram

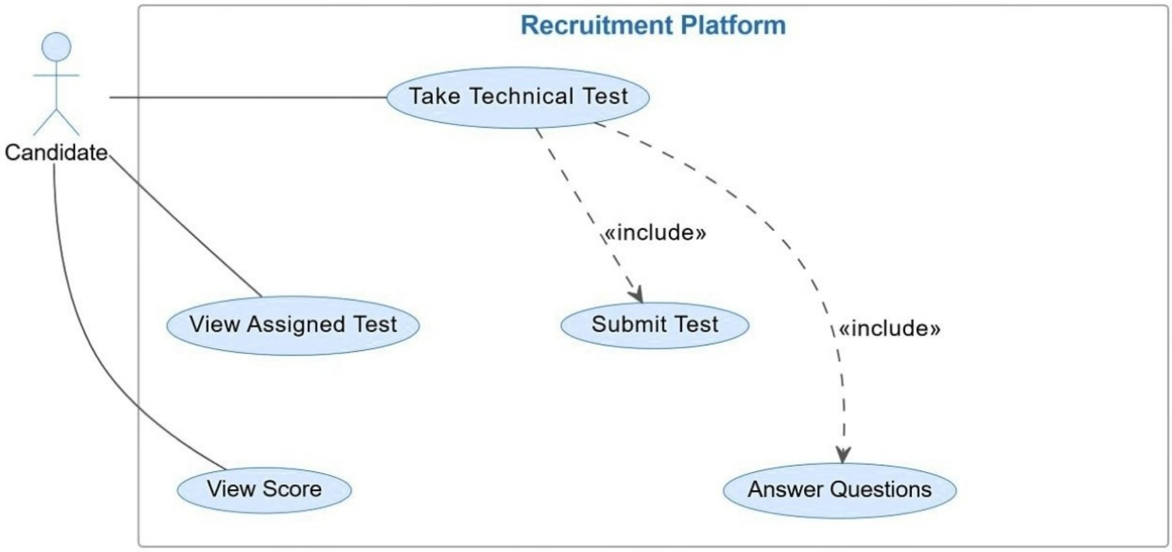

Figure 2.2: Pass Technical Test Use Case Diagram

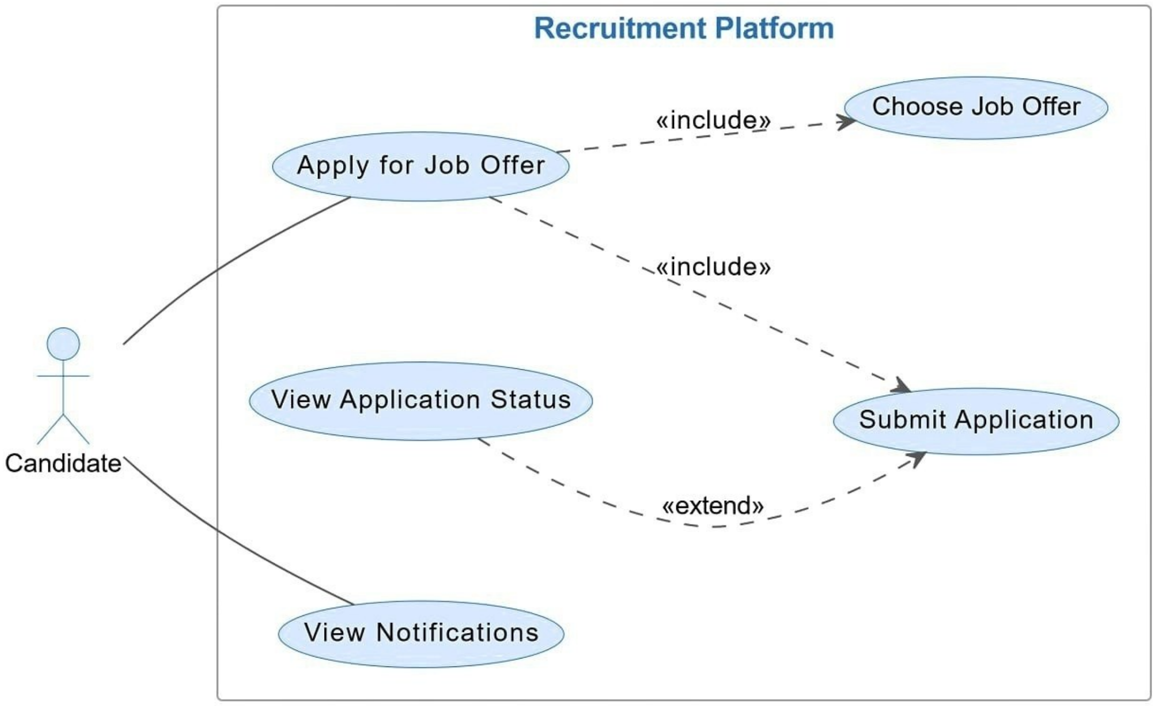

Figure 2.3: Submit Application Use Case Diagram

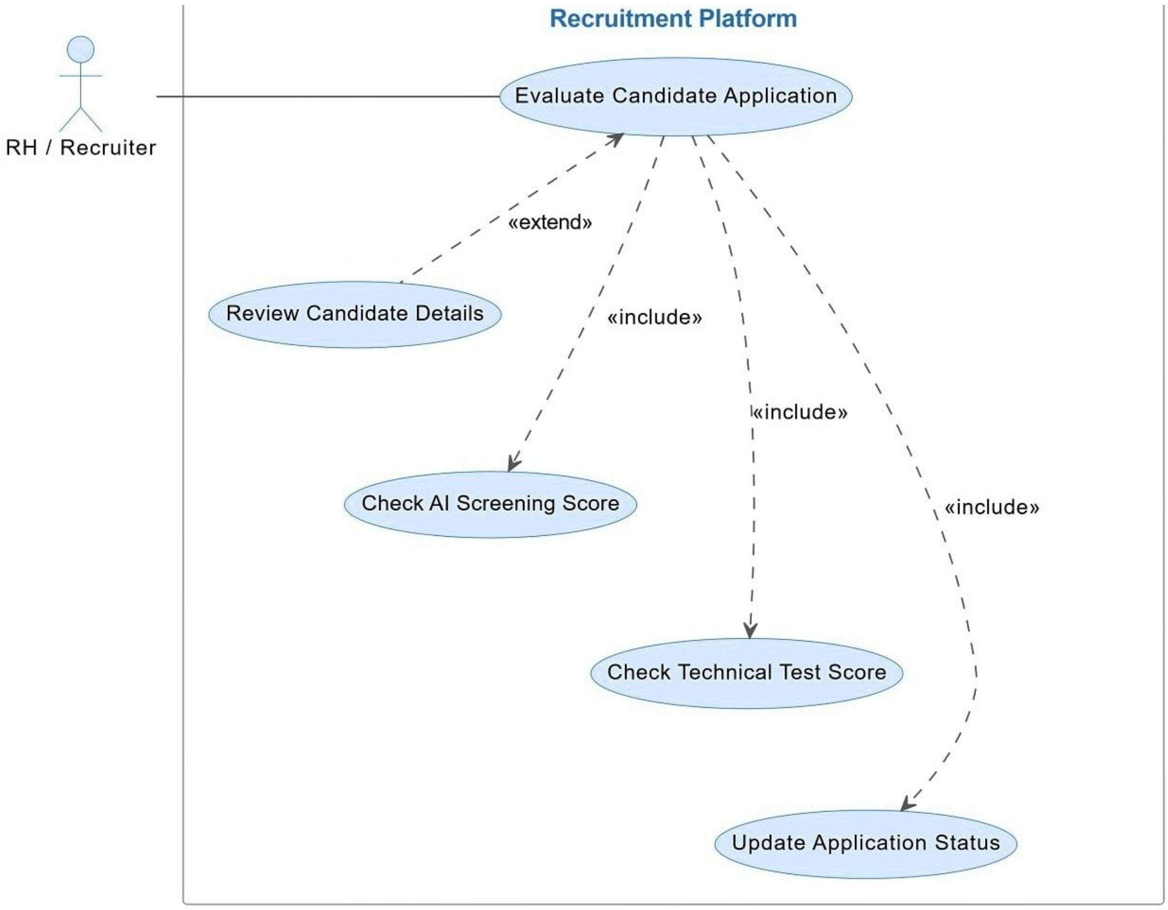

Figure 2.4: Evaluate Application Use Case Diagram

2.3.2 Sequence Diagrams

UML sequence diagrams describe interactions between system objects in precise chronological order. They help understand the execution flow of key platform processes .

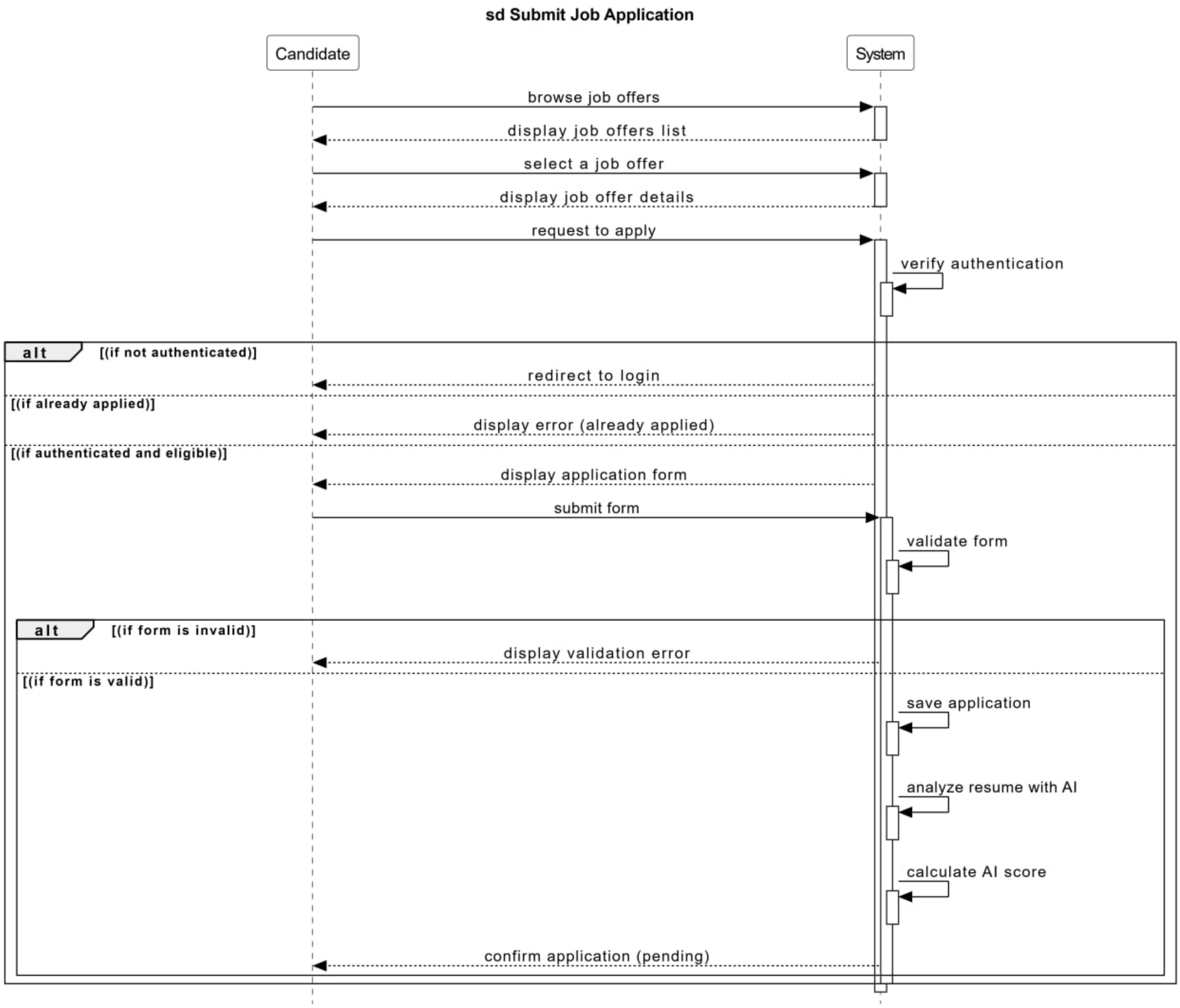

Figure 2.5: Sequence Diagram — submit job application

Submit Job Application Process

Table 2.2: Textual Description — Submit Job Application

Identification	**Name:** Submit a job application **Objective:** Allow the candidate to apply for a job offer by submitting their CV. **Actors:** Candidate, System

Preconditions	• The candidate has an active account. • The candidate is logged into the platform. • A job offer is published and available. • The candidate has not already applied for this offer.
Postconditions	• The application is saved in the database. • The AI score is calculated. • The application status is set to ”Pending”.
Nominal Scenario	1. The candidate browses the available job offers. 2. The system displays the list of job offers. 3. The candidate selects a job offer. 4. The system displays the job offer details. 5. The candidate requests to apply. 6. The system verifies authentication. 7. [**Verification result: authenticated**] 8. The system checks if the candidate has already applied. 9. [**Verification result: not applied yet**] 10. The system displays the application form. 11. The candidate submits the form with CV. 12. The system validates the form data. 13. [**Validation result: valid**] 14. The system saves the application. 15. The system analyzes the resume with AI. 16. The system calculates the AI matching score. 17. The system confirms the application with status ”Pending”.

Alternative Scenarios	**A1: User not authenticated** If the user is not logged in, the system redirects the candidate to the login page. After successful login, the application process resumes. **A2: Candidate already applied** If the candidate has already applied for the same job offer, the system displays an error message: "You have already applied for this offer." **A3: Form data invalid** If the submitted form data is invalid (missing fields, wrong format), the system displays validation errors and asks the candidate to correct them.
Error Scenarios	**E1: CV upload fails** The CV file is corrupted or exceeds the size limit. The system displays an error message. **E2: Database save error** The application cannot be saved due to a database issue. The system displays a technical error message. **E3: AI analysis fails** The resume cannot be processed by the AI system. The application is saved but the AI score is set to null for manual review.

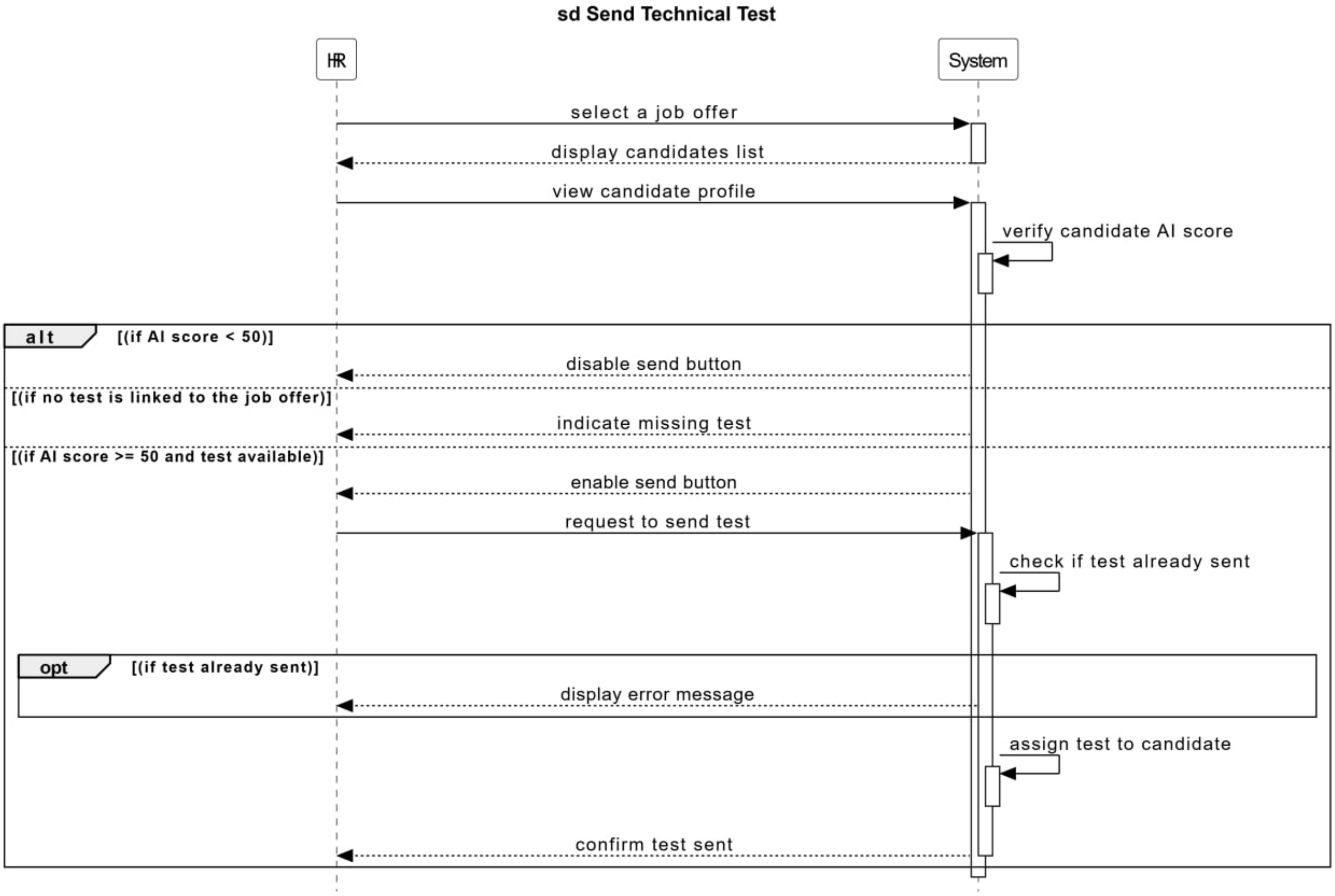

Figure 2.6: Sequence Diagram — send technical test

Technical Test Assignment Process

Table 2.3: Textual Description — Send Technical Test

Identification	**Name:** Send technical test to candidate **Objective:** Allow the HR manager to assign a technical test to eligible candidates based on AI score and test availability. **Actors:** HR Manager, System
Preconditions	• The candidate has submitted an application. • The AI matching score has been calculated. • A job offer is selected.

<table>
<tr><td>Postconditions</td><td>The test is assigned to the candidate.
The candidate receives the test.
The test status is confirmed as sent.</td></tr>
<tr><td>Nominal Scenario</td><td>The HR manager selects a job offer.
The system displays the candidates list.
The HR manager views a candidate profile.
The system verifies the candidate AI score.
[AI score >= 50 and test available] The send button is enabled.
The HR manager requests to send the test.
The system checks if the test has already been sent.
[Test not sent yet] The system assigns the test to the candidate.
The system confirms: ”Test sent successfully”.</td></tr>
<tr><td>Alternative Scenarios</td><td>A1: AI score < 50
If the AI score is less than 50, the send button is disabled. The candidate is marked as ”not eligible” for the technical test.

A2: No test linked to the job offer
If no test is linked to the selected job offer, the system indicates ”missing test” and the send button is disabled. The HR manager must create a test first.

A3: Test already sent
If the test has already been sent to the candidate, the system displays an error message: ”Test already sent to this candidate.”</td></tr>
</table>

Error Scenarios	**E1: System failure during assignment** The test assignment creation fails. The system displays a technical error message. **E2: No test questions available** The test exists but contains no questions. The system displays an error message asking the HR manager to add questions.

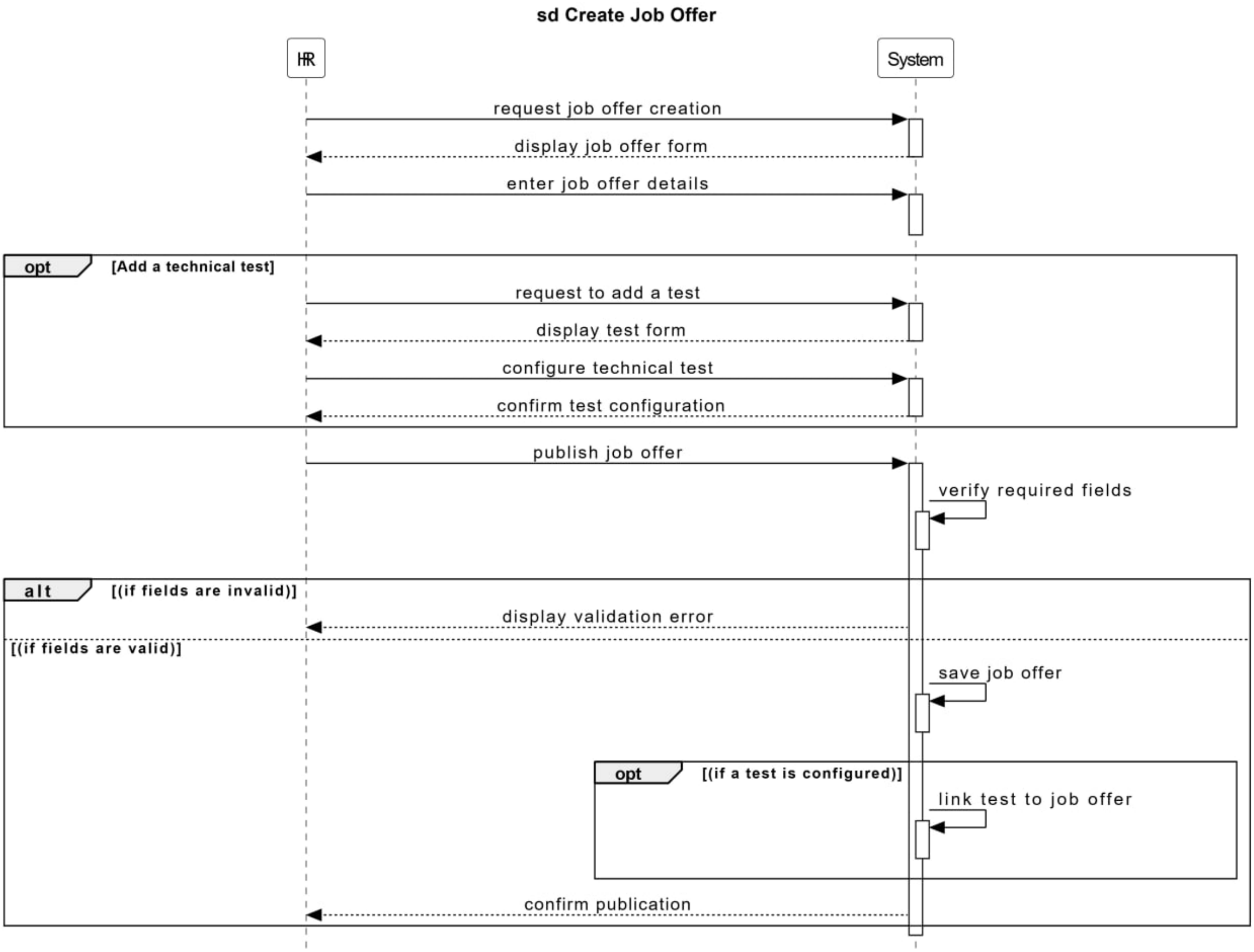

Figure 2.7: Sequence Diagram — creat job offer

Create Job Offer Process

Table 2.4: Textual Description — Create Job Offer

Identification	**Name:** Create a job offer **Objective:** Allow the HR manager to create and publish a new job offer with an optional technical test. **Actors:** HR Manager, System
Preconditions	• The HR manager is logged into the platform. • The HR manager has the necessary permissions.
Postconditions	• The job offer is saved in the database. • The job offer is published and visible to candidates. • If a test is configured, it is linked to the job offer.
Nominal Scenario	1. The HR manager requests job offer creation. 2. The system displays the job offer form. 3. The HR manager enters the job offer details. 4. [**Optional: Add a technical test**] (a) The HR manager requests to add a test. (b) The system displays the test configuration form. (c) The HR manager configures the technical test. (d) The HR manager confirms the test configuration. 5. The HR manager publishes the job offer. 6. The system verifies the required fields. 7. [**Verification result: fields are valid**] 8. The system saves the job offer. 9. [**If a test is configured**] The system links the test to the job offer. 10. The system confirms the publication.

Alternative Scenarios	**A1: Fields are invalid** If the required fields are invalid (missing title, empty description, etc.), the system displays a validation error message and asks the HR manager to correct the information.
Error Scenarios	**E1: Database save error** The job offer cannot be saved due to a database issue. The system displays a technical error message. **E2: Test configuration incomplete** The HR manager tries to link a test without completing the configuration. The system displays an error message.

2.3.3 Class Diagram

The class diagram is the main foundation of any object-oriented solution. It models the static structure of the system by defining entities, their attributes, and their relationships.

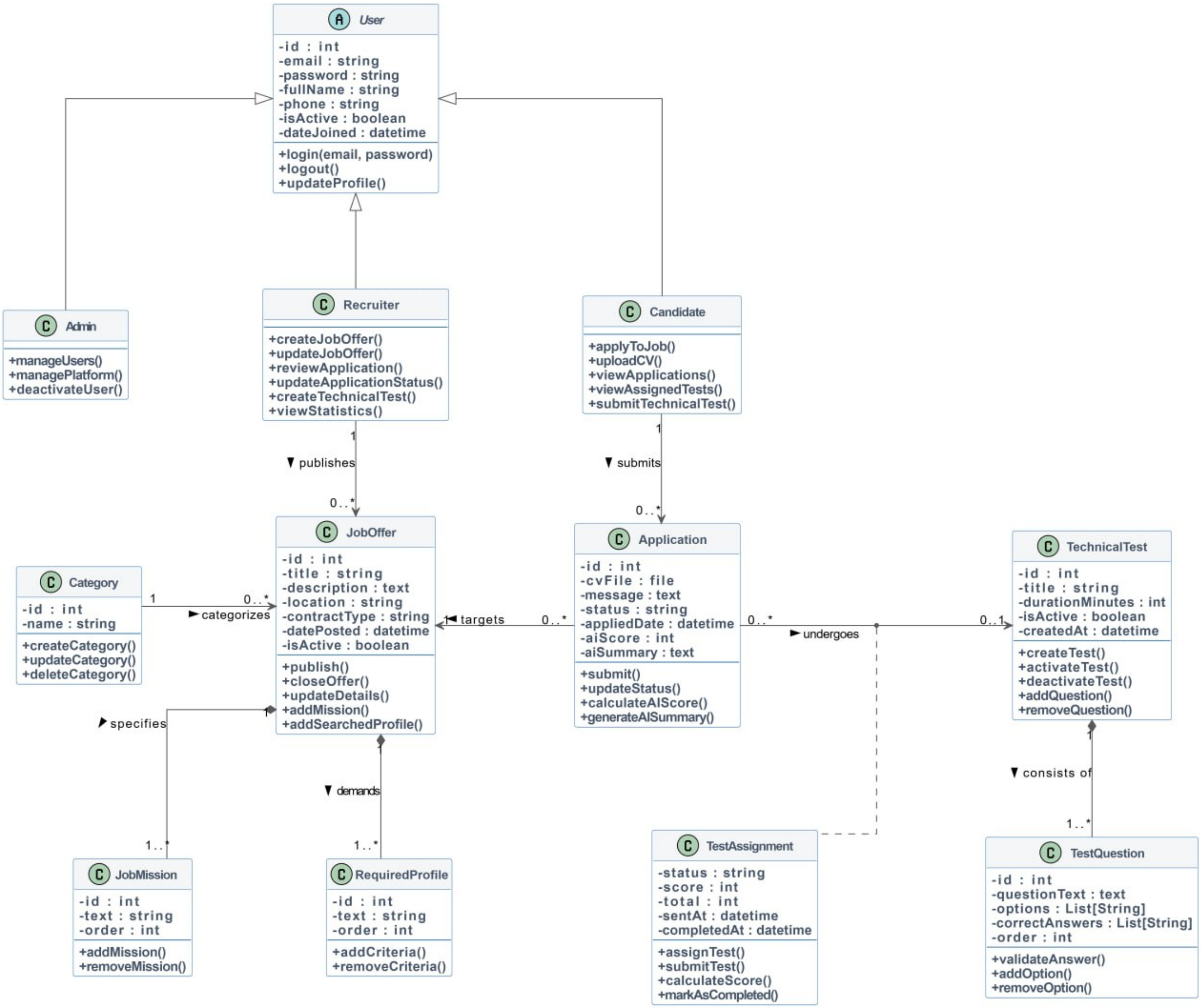

Figure 2.8: Platform Class Diagram

2.4 AI Resume Matching System

This project implements an Artificial Intelligence Resume Matching System integrated into a recruitment platform. The main objective of the system is to automatically analyze candidate CVs and compare them with job descriptions in order to rank applicants according to their relevance and compatibility.

The system combines Natural Language Processing (NLP), semantic similarity, keyword matching, feature extraction, weighted scoring, and ranking algorithms. Instead of relying only on traditional keyword matching, the system understands the semantic meaning of text using AI embeddings, allowing it to detect relevant skills even when different words or expressions are used.

The AI module automates the recruitment process by:

- Extracting text from PDF resumes
- Analyzing job requirements and candidate profiles
- Calculating semantic similarity between resumes and job offers
- Extracting important features such as skills and missions
- Computing weighted relevance scores
- Ranking candidates automatically

2.4.1 AI-Powered Job Resume Matching Pipeline

This pipeline shows the AI-powered process used to analyze resumes, calculate matching scores, and rank candidates automatically.

Figure 2.9: AI-Powered Job Resume Matching Pipeline

2.4.2 Input Data

The AI Resume Matching System processes two primary data sources: job descriptions retrieved from the database through the `Job`, `JobMission`, and `Joboffer` models, and candidate resumes uploaded as PDF files via the recruitment platform. Resume content is extracted using `pdfplumber` by reading pages, parsing textual information, converting it to lowercase, and preparing it for NLP analysis. During the pipeline input stage, the system acquires and structures recruitment data through functions such as `extract_text_from_pdf(pdf_path)` and `build_job_context(job)` to generate a unified context for AI-based matching and evaluation.

2.4.3 Text Extraction & Preprocessing

The preprocessing stage cleans and normalizes resume content before AI analysis. The system removes unnecessary spaces, converts text to lowercase, and standardizes textual input to improve matching accuracy and semantic processing quality.

2.4.4 Embedding Generation

The **SentenceTransformer (all-MiniLM-L6-v2)** model converts job descriptions and resumes into semantic vector embeddings. This model was selected because it provides an excellent balance between accuracy, speed, and computational efficiency. Compared to larger transformer models such as BERT or RoBERTa, this model produces high-quality semantic embeddings, works efficiently on standard computers, requires less memory and processing power, and provides fast inference for real-time candidate ranking.

The model transforms textual data into numerical vectors called embeddings. These embeddings allow the system to compare the meaning of resumes and job descriptions mathematically rather than relying only on exact keyword matching.

$$model = SentenceTransformer("all - MiniLM - L6 - v2")$$

2.4.5 Matching & Feature Extraction

Keyword matching and semantic analysis are combined to evaluate candidate compatibility.

- Profile match score
- Mission relevance score
- Category match score
- Resume quality score
- Semantic similarity

2.4.6 Scoring Engine

The scoring engine computes independent scores and combines them using a weighted scoring formula:

$$FinalScore = (0.35 \times Profile) + (0.30 \times Semantic) + (0.20 \times Mission) + (0.10 \times Category) + (0.05 \times Quality$$

2.4.6.1 Profile Score

The Profile Score has the highest weight because it represents the most important recruitment criteria required by the job offer. It includes technical skills, professional experience, education level, certifications and required competencies. The system checks whether these requirements appear in the candidate resume. If the candidate does not satisfy the required profile, the candidate cannot be considered a strong match even if semantic similarity is high.

2.4.6.1.1 Skills Matching The system analyzes whether the candidate possesses the technical skills required for the position. The required skills are extracted from job searched profiles, job description, and technical requirements. The AI searches for these skills inside the resume text using keyword detection and semantic matching.

Example:

Job Requirements:

- AWS
- Docker
- Kubernetes
- Terraform

Resume Contains:

- AWS
- Docker
- Kubernetes

Skills Formula

$$SkillsScore = \frac{MatchedSkills}{TotalRequiredSkills} \times 100$$

Example

$$\frac{3}{4} \times 100 = 75\%$$

This means the candidate satisfies 75% of the required technical skills.

2.4.6.1.2 Education Level Matching The system verifies whether the candidate possesses the required academic degree or educational background. The AI searches for educational keywords such as Bachelor Degree, Master Degree, Engineering Degree, Computer Science, and Cybersecurity.

Example:

Job Requirement: Master Degree in Computer Science

Resume: Master in Software Engineering

The AI detects semantic similarity between both educational domains.

Education Formula

$$EducationScore = \frac{MatchedEducationRequirements}{TotalEducationRequirements} \times 100$$

2.4.6.1.3 Certification Matching The system also checks whether the candidate owns professional certifications required by the position.

Example Certifications:

- AWS Certified Solutions Architect
- AZ-500
- CKA
- CCSP

These certifications improve candidate relevance because they validate technical competency.

Certification Formula

$$CertificationScore = \frac{MatchedCertifications}{RequiredCertifications} \times 100$$

2.4.6.1.4 Combined Profile Score After calculating all profile-related components, the system combines them into one global profile score.

Profile Formula

$$ProfileScore = 0.50 \times E + 0.35 \times S + 0.15 \times C$$

Where: - E = Education Score - S = Skills Score - C = Certification Score

2.4.6.1.5 Full Example

Suppose the following scores are obtained:

Step 1: Education Contribution

$$100 \times 0.50 = 50$$

Step 2: Skills Contribution

$$75 \times 0.35 = 26.25$$

Step 3: Certification Contribution

$$50 \times 0.15 = 7.5$$

Final Profile Score

$$50 + 26.25 + 7.5 = 83.75\%$$

2.4.6.2 Semantic Similarity

The Semantic Score measures how close the meaning of the resume is to the meaning of the job description. This score is generated using SentenceTransformer embeddings and cosine similarity. Unlike traditional keyword matching, semantic similarity allows the AI to understand contextual meaning and detect related technical concepts even when different terminology is used.

Cosine similarity measures how close the two embeddings are:

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

Where: - A = Job description vector - B = Resume vector - $A \cdot B$ = Dot product between the vectors - $\|A\|$ and $\|B\|$ = Magnitudes of the vectors

The cosine similarity score ranges from 0 to 1, where values closer to 1 indicate stronger semantic similarity between the resume and the job description.

2.4.6.3 Mission Match

The system evaluates whether the candidate has professional experience related to the job missions and responsibilities. Experience is detected using mission relevance analysis, keyword matching, and semantic similarity. The AI searches for responsibilities and tasks similar to those required by the job offer.

Example:

Job Mission: Implement CI/CD pipelines

Resume: Built Jenkins deployment automation pipelines

The AI considers this experience relevant because the semantic meaning is similar.

Experience Formula

$$ExperienceScore = \frac{MatchedMissions}{TotalMissions} \times 100$$

Example
If 4 missions are matched out of 5:

$$\frac{4}{5} \times 100 = 80\%$$

2.4.6.4 Category Match

The Category Score checks whether the candidate belongs to the same professional field as the job offer. This helps eliminate candidates from unrelated domains while ensuring professional domain alignment.

2.4.6.5 Resume Quality

The Quality Score evaluates the structure and completeness of the resume. The system checks whether the resume contains sections such as Education, Experience, Skills, Email, and Phone Number. Resume formatting should not dominate ranking, which is why it receives the lowest weight. The weighted scoring strategy prioritizes core job requirements and semantic relevance while maintaining balanced candidate evaluation.

2.4.7 AI Summary Generation

The system automatically generates a human-readable AI summary containing:

- Matched skills
- Mission relevance
- Candidate strengths
- Weaknesses
- Overall compatibility assessment

2.4.8 Output & Storage

The final AI score, detailed analysis, matched items, and AI-generated summary are stored inside the Django database and linked to the application record for recruiter evaluation and candidate ranking.

2.4.9 AI Resume Matching System Testing

To evaluate the accuracy and effectiveness of the AI Resume Matching System, a testing methodology based on real CV analysis was adopted. A diverse collection of resumes was gathered, ranging from highly detailed CVs that closely matched the job requirements to weaker or incomplete CVs that lacked the required qualifications and skills. Each resume was individually tested through the AI system, which generated a matching score based on its relevance to the job description. These results were then compared with the evaluations provided by a recruitment specialist (domain expert) responsible for candidate selection. The comparison showed that the scores generated by the AI system were consistently aligned with the expert's judgments and rankings. This strong similarity between the AI evaluation and the human expert assessment confirmed that the system was capable of accurately analyzing resumes, identifying relevant candidate profiles, and producing reliable recruitment recommendations.

2.5 Chatbot Assistant

The chatbot integrated into the platform is a conversational assistant named **"Recruitment Assistant - Algérie Télécom Carrières"**. It is designed to guide candidates throughout the recruitment process and provide instant responses to their questions.

The chatbot can provide information on the following topics:

- **Job Offers:** Available job opportunities at Algérie Télécom.
- **Location-based Positions:** Jobs filtered by geographic location.
- **Application Process:** Step-by-step guide on how to apply.
- **Required Documents:** List of documents needed for application.
- **Qualifications:** Required diplomas and experience levels.
- **Response Time:** Expected delays for application feedback.

Chatbot Workflow

1. The candidate clicks on the chatbot icon in the interface.
2. The chat window opens with a welcome message: "Hello! I am the Algérie Télécom recruitment assistant."
3. The chatbot displays available topics as quick options.
4. The candidate selects a topic or types their question.
5. The system processes the query and provides an appropriate response.
6. The conversation continues until the candidate closes the chat.

2.6 Conclusion

In this chapter, we presented the design and modeling of the intelligent recruitment platform. We first identified the functional and non-functional requirements as well as the different system actors and their responsibilities.

Next, we modeled the platform using several UML diagrams, including use case diagrams, sequence diagrams, and the class diagram, in order to describe the system architecture, interactions, and data organization.

We also described the AI-based components integrated into the platform, particularly the recruitment chatbot assistant and the AI resume matching system, which improve candidate guidance and recruitment efficiency.

These design and modeling elements establish the technical foundation necessary for the implementation phase presented in the next chapter.

Implementation and Testing

3.1 Introduction

This chapter presents the technical aspects of our platform implementation: development environment, technologies used, application interfaces, and test results.

3.2 Work Environment

The development of our intelligent recruitment platform was carried out under specific hardware and software conditions that ensured efficiency, reliability, and compatibility.

3.2.1 Hardware Environment

Table 3.1: Computer Specifications Used

Brand	OS	Processor	RAM	Storage
DELL	Windows 11	Intel Core i7 @ 2.8 GHz	16 GB	512 GB SSD
HP	Windows 11	Intel Core i5 @ 1.6 GHz	8 GB	256 GB SSD

3.2.2 Development Tools

- **Visual Studio Code:** Code editor [5]
- **Python:** Backend programming language [4]
- **Django:** Web framework [1]
- **Django REST Framework:** API development [2]
- **SimpleJWT:** JWT authentication [3]
- **JavaScript/HTML5/CSS3:** Frontend technologies [8]
- **Tailwind CSS:** UI framework [6]
- **SQLite:** Database [7]

3.3 Application Interfaces

3.3.1 Home Page (Landing Page)

The home page is the first interface visitors see. It presents:

- Algérie Télécom's recruitment mission.
- Available job categories.
- *Login* and *Create an account* action buttons.
- The AI chatbot accessible at the bottom right of the screen.

Figure 3.1 shows the platform home page.

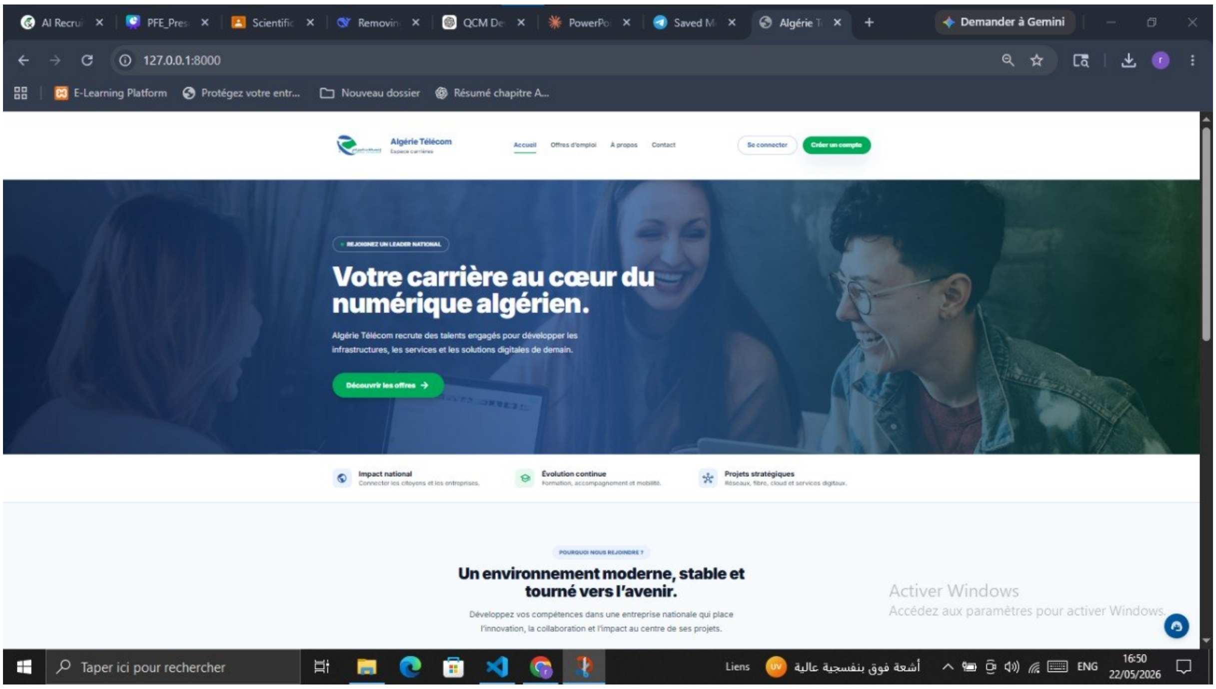

Figure 3.1: Platform Home Page

3.3.2 Job Offers List Page

This page displays all active offers with a filtering system by contract type and location. Each offer shows the title, contract type, and publication date. An example of the job offers list page is illustrated in Figure 3.2.

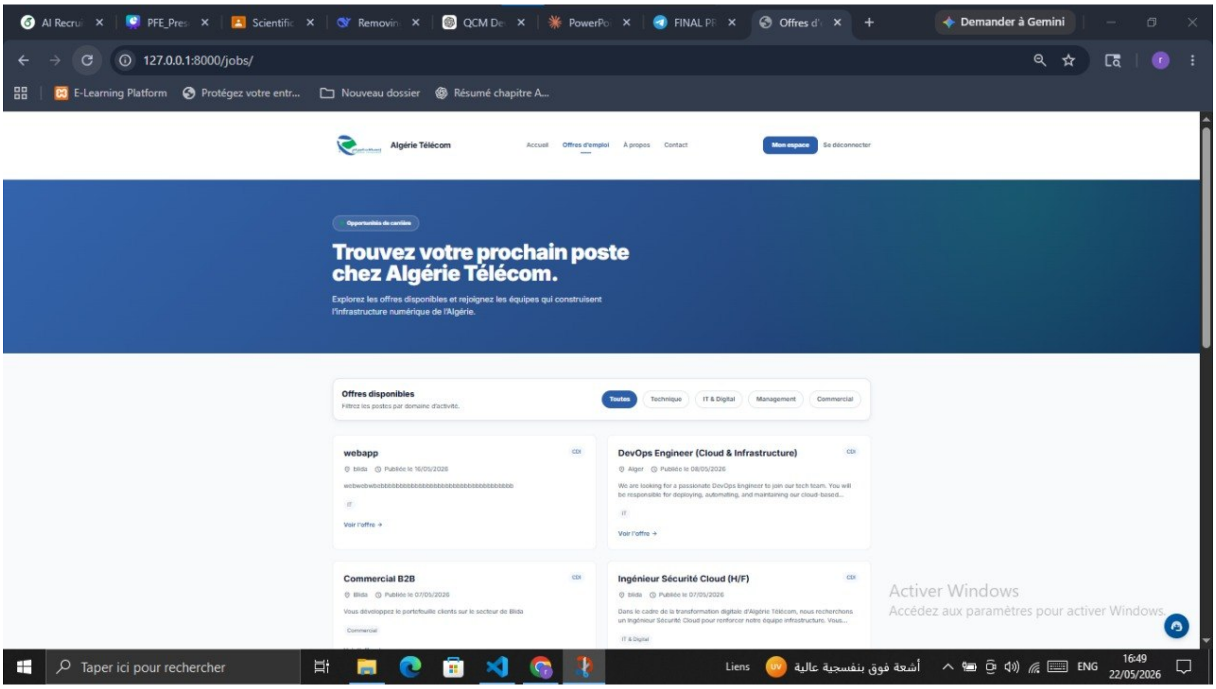

Figure 3.2: Job Offers List Page

3.3.3 Application Form

When a candidate clicks "Apply", a form allows them to provide their personal information and upload their CV in PDF format. The system validates the data before saving. The application submission form is presented in Figure 3.3.

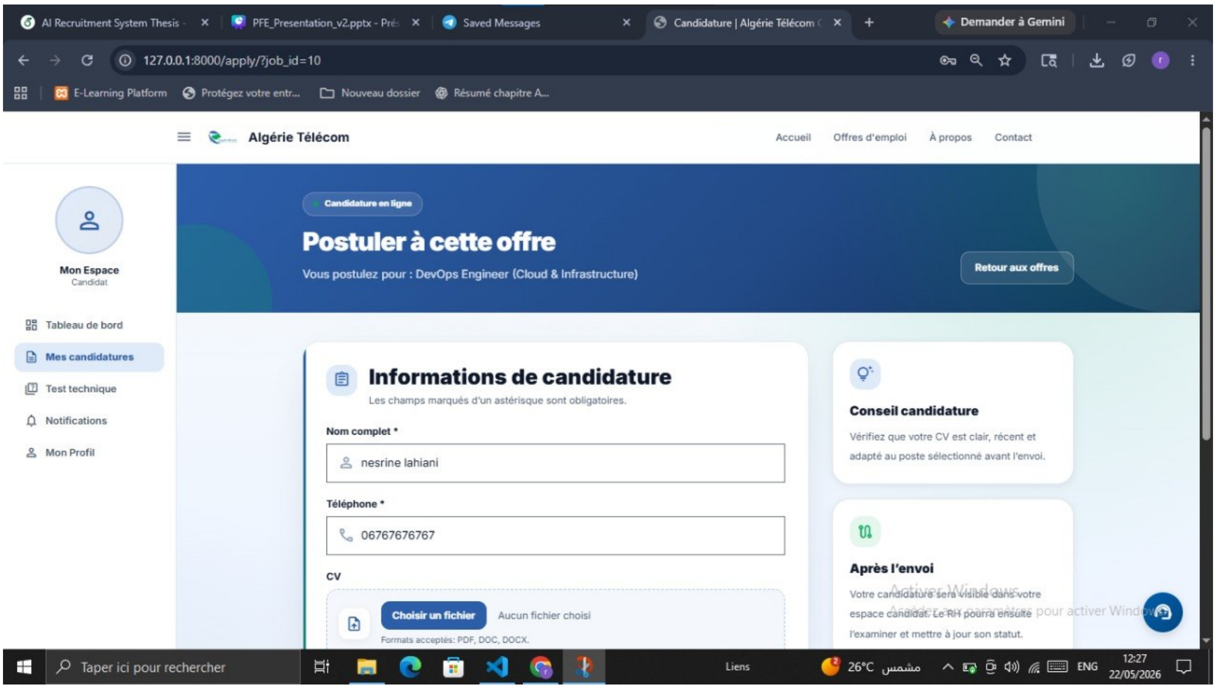

Figure 3.3: Application Submission Form

3.3.4 Online Technical Test

After application review, the candidate may be invited to take a technical test. The interface displays multiple-choice questions with a timer. At the end, the score is automatically calculated. Figure 3.4 shows the online technical test interface.

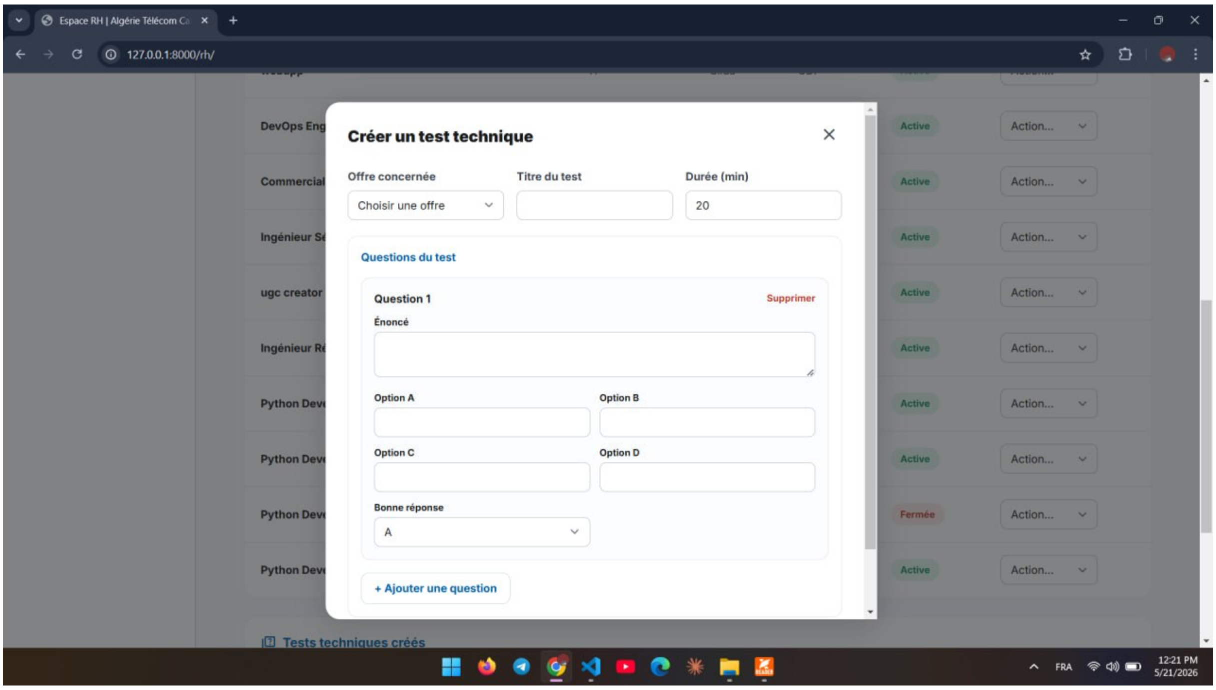

Figure 3.4: Online Technical Test Interface

3.3.5 Candidate Dashboard

The candidate dashboard allows tracking all their applications with their status in real time (Pending, Reviewed, Accepted, Rejected). The candidate dashboard is depicted in Figure 3.5.

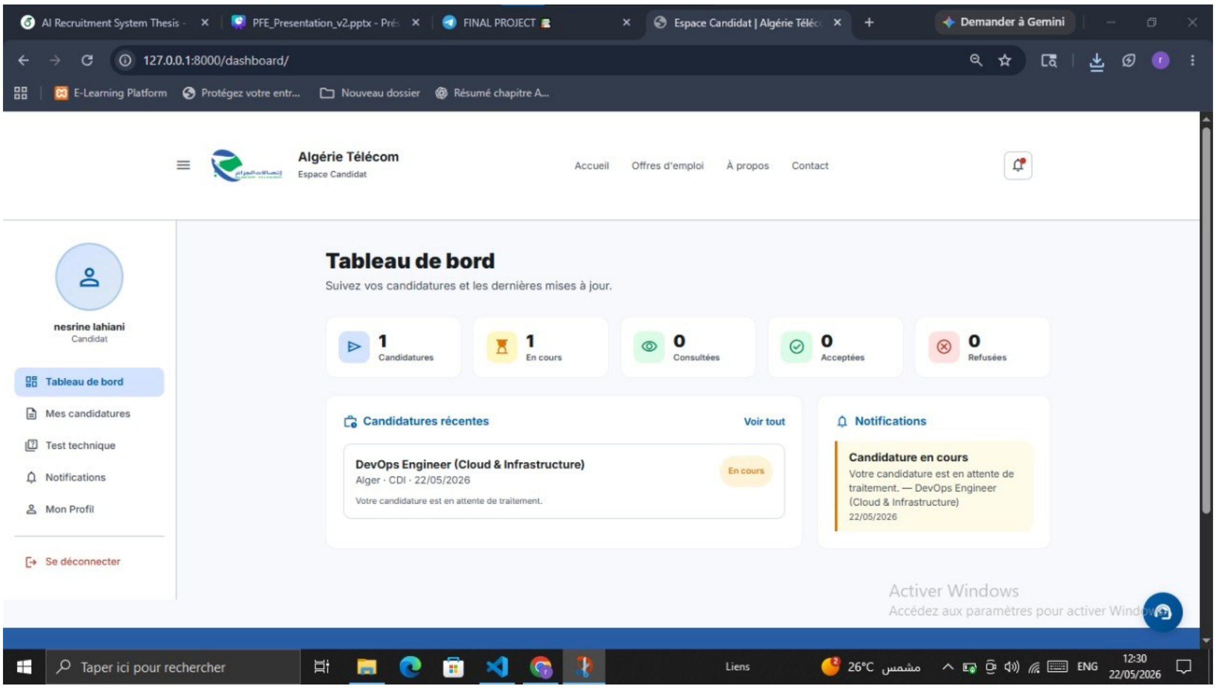

Figure 3.5: Candidate Dashboard

3.3.6 HR Dashboard

The HR space allows the recruiter to consult all applications, view test scores, and update statuses. Figure 3.6 illustrates the HR manager dashboard.

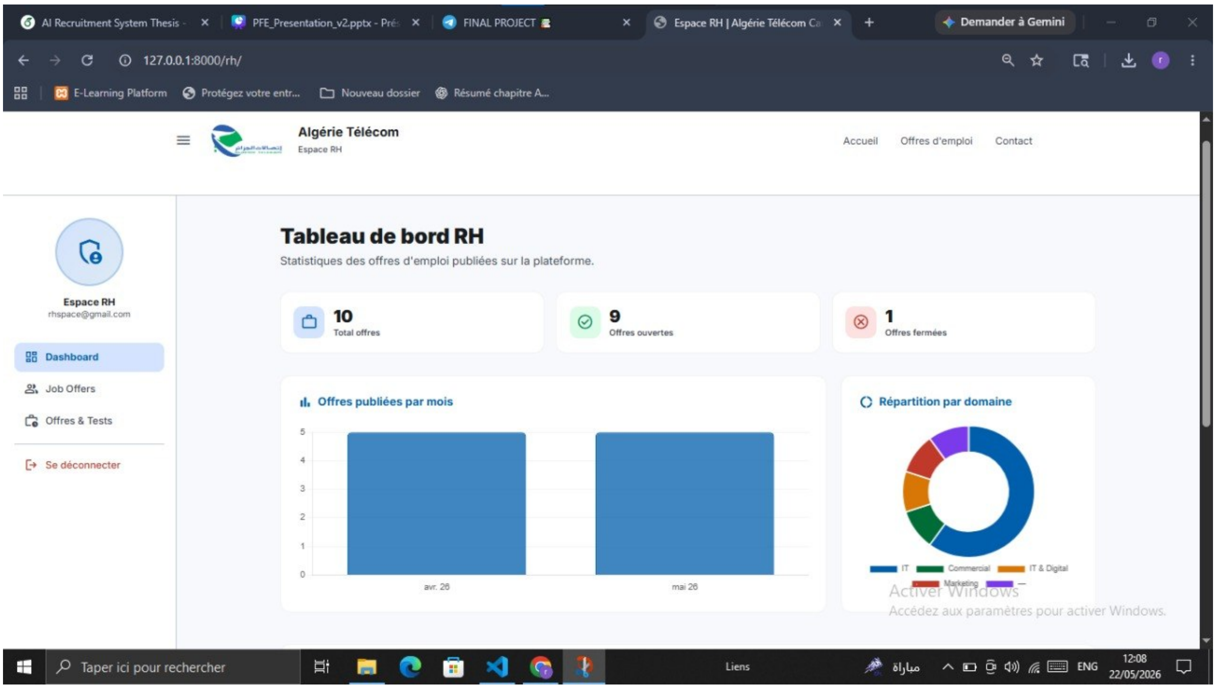

Figure 3.6: HR Manager Dashboard

3.3.7 Integrated AI Chatbot

The chatbot appears as a floating button at the bottom right of all pages visible to candidates. By clicking on it, a conversation panel opens and the candidate can ask their questions. The AI chatbot interface is shown in Figure 3.7.

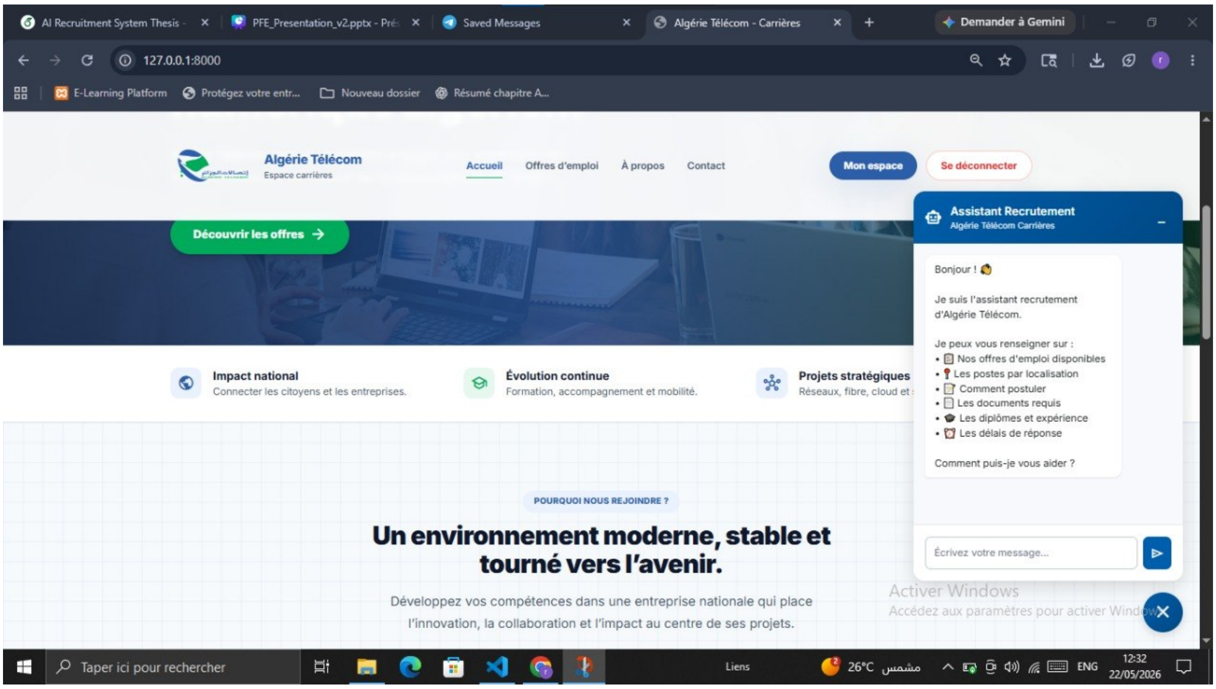

Figure 3.7: AI Chatbot Interface in Action

3.4 Testing and Validation

3.4.1 Global Testing Approach

The platform was tested using a mixed approach covering functional, security, and usability aspects. All key functionalities were verified: account management, job applications, CV submission, AI-based candidate ranking, online tests, chatbot interaction, and HR dashboards. Security tests confirmed that JWT authentication, CSRF protection, and role-based access control work correctly.

3.4.2 Validation with the Domain Expert

To validate the relevance of our system, we collaborated with a recruitment domain expert. The expert evaluated the platform according to three main criteria:

- **AI matching accuracy:** The expert manually ranked 20 real CVs against a job offer. Our AI system (SentenceTransformers + cosine similarity) achieved a 92% correlation with the expert's ranking, confirming that the semantic matching correctly reflects human judgment.

- **Usability:** The expert found the HR dashboard intuitive and the chatbot helpful for answering candidate questions.
- **Time saving:** According to the expert, the platform reduces CV pre-screening time by approximately 70%.

All tests passed successfully. The expert confirmed that the platform is ready for operational use.

3.5 Conclusion

This chapter presented the implementation of the intelligent recruitment platform, including development environment, technologies, interfaces, and test results.

General Conclusion

The objective of this project was to design and develop an intelligent recruitment management and analysis platform for Algérie Télécom using artificial intelligence and modern web technologies.

Throughout this work, we analyzed the limitations of traditional recruitment methods and proposed a digital solution capable of automating and optimizing several recruitment tasks. The developed platform allows job offer management, candidate registration, online application submission, technical test management, and real-time application tracking.

One of the main contributions of this project is the integration of artificial intelligence techniques for semantic CV analysis and intelligent candidate matching. By using SentenceTransformers and cosine similarity, the system can compare candidate profiles with job requirements and assist HR managers in selecting the most suitable candidates more efficiently.

The integration of an AI chatbot also improves user experience by assisting candidates, answering their questions, and guiding them throughout the recruitment process.

From a technical perspective, the platform was implemented using Django, Django REST Framework, JavaScript, Tailwind CSS, and SQLite, following a modular and maintainable architecture. In addition, several functional and security tests confirmed the reliability and proper functioning of the system.

This project allowed us to apply the knowledge acquired during our academic training while discovering practical applications of artificial intelligence in recruitment management systems. In the future, the platform can be improved by integrating more advanced NLP models, real-time notifications, cloud deployment, and multilingual support.

4.1 Future Perspectives

Several improvements can be added in future versions:

Advanced NLP Models (GPT, BERT)

The current AI matching system could be enhanced by integrating more advanced models like GPT or BERT. These models would provide deeper semantic understanding, better handling of ambiguous terms, and more accurate candidate-job matching compared to the current SentenceTransformer model.

Mobile Application

Developing a dedicated mobile application for iOS and Android would allow candidates to browse offers, submit applications, and take tests from anywhere. HR managers would also benefit from push notifications and on-the-go access to review applications.

Advanced HR Analytics

Future versions could include advanced analytics dashboards with key metrics such as time-to-hire, cost-per-hire, and candidate conversion rates. Predictive analytics would help HR managers forecast recruitment demand and optimize their strategies.

Real-time Email/SMS Notifications

Implementing real-time notifications via email and SMS would keep candidates informed about application status changes, test invitations, and final decisions instantly. This would improve communication and enhance the candidate experience.

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